

## SCHWERTDFEGER, Antarctica

WMO: 898680

Lat: 79.816S

Lon: 170.358E

Elev: 177

StdP: 14.60

Time Zone: 12.00 (NZL)

Period: 94-12

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-----|-----|------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99% |     |      | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 8             | -70.3      | -65.7 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 39.3                  | -24.1 | 33.7 | -23.7 | 5.6                   | 190  | 1.400 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                     | (p)  |
| 12            | 13.6                      | 30.2            | N/A  | 27.6 | N/A  | 25.7 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 6.8                     | 350  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 31.7          | 27.3 | 23.8 | DB                         | -74.2 | 39.2               | 5.6 | 3.2                                                | -78.2 | 41.5       | -81.5 | 43.3       | -84.6 | 45.1       | -88.7 | 47.4 | (4) |
| (5)               |      |      | WB                         | N/A   | N/A                | N/A | N/A                                                | N/A   | N/A        | N/A   | N/A        | N/A   | N/A        | N/A   | N/A  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -15.6  | 17.0  | -0.3  | -19.5 | -25.5 | -29.5 | -30.8 | -33.3 | -38.3 | -31.4 | -18.0 | 4.3   | 18.4  | (6)  |  |
|                                                                        | DBStd       | 22.80  | 5.69  | 9.09  | 10.76 | 12.73 | 15.45 | 15.03 | 15.97 | 14.08 | 13.90 | 10.51 | 8.98  | 5.43  | (7)  |  |
|                                                                        | HDD50       | 23958  | 1022  | 1408  | 2155  | 2266  | 2466  | 2423  | 2583  | 2738  | 2442  | 2107  | 1370  | 978   | (8)  |  |
|                                                                        | HDD65       | 29433  | 1487  | 1828  | 2620  | 2716  | 2931  | 2873  | 3048  | 3203  | 2892  | 2572  | 1820  | 1443  | (9)  |  |
|                                                                        | CDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (10) |  |
|                                                                        | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (13) |  |
|                                                                        | WSAvg       | 10.3   | 7.3   | 7.8   | 9.6   | 10.7  | 11.5  | 11.3  | 12.3  | 11.5  | 12.7  | 11.7  | 10.2  | 7.2   | (14) |  |
|                                                                        | PrecAvg     | 6.7    | 0.6   | 0.7   | 0.7   | 0.6   | 0.6   | 0.6   | 0.5   | 0.6   | 0.6   | 0.4   | 0.4   | 0.5   | (15) |  |
| (16) Precipitation                                                     | PrecMax     | 10.9   | 1.4   | 1.4   | 1.4   | 1.5   | 1.6   | 1.6   | 1.4   | 1.7   | 1.5   | 1.3   | 1.1   | 1.0   | (16) |  |
|                                                                        | PrecMin     | 3.1    | 0.2   | 0.1   | 0.1   | 0.1   | 0.0   | 0.1   | 0.1   | 0.0   | 0.0   | 0.1   | 0.0   | 0.1   | (17) |  |
|                                                                        | PrecStd     | 1.9    | 0.3   | 0.3   | 0.4   | 0.4   | 0.4   | 0.4   | 0.3   | 0.4   | 0.4   | 0.3   | 0.3   | 0.2   | (18) |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |      |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 33.4  | 25.0  | 10.3  | 17.0  | 9.7   | 9.8   | 8.2   | 3.4   | 9.9   | 8.5   | 26.6  | 36.4  | (19) |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (20) |  |
|                                                                        | 2%          | DB     | 29.4  | 19.3  | 4.7   | 5.3   | 2.2   | 3.1   | 1.3   | -3.8  | 3.4   | 5.0   | 23.2  | 31.6  | (21) |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (22) |  |
|                                                                        | 5%          | DB     | 27.3  | 16.2  | 0.7   | -2.1  | -3.4  | -4.5  | -5.7  | -11.0 | -4.4  | 1.4   | 20.2  | 28.8  | (23) |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (24) |  |
|                                                                        | 10%         | DB     | 25.7  | 13.5  | -3.2  | -6.8  | -8.2  | -9.5  | -11.1 | -17.7 | -12.0 | -2.3  | 17.1  | 26.5  | (25) |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (26) |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (27) |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (28) |  |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (29) |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (30) |  |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (31) |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (32) |  |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (33) |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (34) |  |
| (36) Mean Daily Temperature Range                                      |             | MDBR   | 12.4  | 16.4  | 16.0  | 16.7  | 17.2  | 17.5  | 17.0  | 18.3  | 17.8  | 17.3  | 14.8  | 13.6  | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 15.1  | 17.4  | 16.9  | 21.9  | 21.6  | 20.7  | 19.2  | 23.0  | 21.6  | 19.9  | 16.0  | 17.4  | (36) |  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (38) |  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.233  | 0.211 | 0.163 | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | 0.155 | 0.192 | 0.212 | 0.231 | (40) |  |
|                                                                        | taud        | 2.182  | 2.191 | 2.100 | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | 2.056 | 2.182 | 2.204 | 2.187 | (41) |  |
|                                                                        | Ebn at Noon | 301    | 278   | 227   | 0     | 0     | 0     | 0     | 0     | 0     | 226   | 288   | 310   | 310   | (42) |  |
|                                                                        | Edh at Noon | 32     | 24    | 13    | 0     | 0     | 0     | 0     | 0     | 0     | 13    | 24    | 31    | 34    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 2552   | 1508  | 441   | 0     | 0     | 0     | 0     | 0     | 0     | 211   | 1105  | 2421  | 3003  | (44) |  |
|                                                                        | RadStd      | 73     | 63    | 21    | 0     | 0     | 0     | 0     | 0     | 0     | 11    | 32    | 79    | 80    | (45) |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | N/A   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page